

OOP Activity 3

Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

Name: Adhiraj Kale

Division: SY2 ETC

PRN: B25ET1093

CODE

```
#include <iostream>
using namespace std;
void inputArray(int arr[], int size){
    for(int i=0; i<size; i++){
        cin >> arr[i];
    }
}
void outputArray(int arr[], int size){
    for(int i=0; i<size; i++){
        cout << arr[i] << " ";
    }
}
void swap(int &a, int &b){
    int temp = a;
    a = b;
    b = temp;
}
void ascending(int arr[], int size){
    for (int i = 0; i < size - 1; i++) {
        for (int j = 0; j < size - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                swap(arr[j],arr[j+1]);
            }
        }
    }
}

void descending(int arr[], int size){
    for (int i = 0; i < size - 1; i++) {
        for (int j = 0; j < size - i - 1; j++) {
            if (arr[j] < arr[j + 1]) {
                swap(arr[j],arr[j+1]);
            }
        }
    }
}

int main() {
    int arr[10], size;
    cout << "Enter the size of the array: ";
    cin >> size;

    cout << "Enter an array: ";
    inputArray(arr, size);

    cout << "The array you entered is: ";
    outputArray(arr, size);

    cout << endl;
    cout << "Ascending order is: ";
    ascending(arr, size);
    for(int i=0; i<size; i++){
        cout << arr[i] << " ";
    }
}
```

```
}

cout << endl;
cout << "Descending order is: ";
descending(arr, size);
for(int i=0; i<size; i++){
    cout << arr[i] << " ";
}

return 0;
}
```

OUTPUT

Output

```
Enter the size of the array: 4
Enter an array: 3
5
6
4
The array you entered is: 3 5 6 4
Ascending order is: 3 4 5 6
Descending order is: 6 5 4 3
```